

INARI VAYLA, Finland

WMO: 028270

Lat: 69.0722N

Lon: 27.4925E

Elev: 405

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.4	-23.6	-35.1	0.8	-28.3	-29.7	1.1	-23.4	25.7	22.8	22.8	22.4	3.1	320	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	75.3	62.4	71.1	60.1	67.3	58.4	63.9	72.5	61.9	69.2	59.6	65.8	10.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	80.6	67.0	58.5	74.2	64.6	56.6	69.3	62.6	29.4	72.6	27.9	69.2	26.3	66.1	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.1	15.8	DB	-36.1	80.1	4.4	4.1	-39.2	83.0	-41.8	85.4	-44.3	87.7	-47.5	90.6	(4)
(5)			WB	-35.0	66.2	4.9	3.8	-38.5	68.9	-41.4	71.1	-44.1	73.2	-47.6	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	7.0	8.4	16.8	29.0	40.8	49.4	56.2	51.6	44.8	31.9	20.3	13.6		
	DBStd	19.50	14.22	15.36	11.02	8.24	7.26	7.16	7.06	6.21	5.75	7.68	10.57	12.73		
	HDD50	7382	1332	1164	1031	631	301	97	20	53	175	560	890	1128		
	HDD65	12446	1797	1584	1496	1081	749	471	285	418	607	1025	1340	1593		
	CDD50	427	0	0	0	0	17	78	211	103	18	0	0	0		
	CDD65	18	0	0	0	0	0	3	12	3	0	0	0	0		
	CDH74	209	0	0	0	0	9	24	148	28	0	0	0	0		
	CDH80	45	0	0	0	0	1	3	34	7	0	0	0	0		
	WSAvg	7.2	6.6	6.5	8.3	7.3	7.6	8.1	7.2	6.7	7.2	7.2	6.8	7.0		
	PrecAvg	17.7	1.1	0.9	0.9	0.9	1.2	2.1	2.7	2.3	1.6	1.7	1.2	1.2		
	PrecMax	23.4	2.0	2.4	1.6	2.5	2.8	4.3	5.4	6.5	3.6	3.9	2.5	2.1		
(15) Precipitation	PrecMin	10.9	0.5	0.2	0.4	0.2	0.4	0.2	0.9	0.9	0.3	0.3	0.4	0.6		
	PrecStd	2.9	0.4	0.5	0.4	0.5	0.6	1.0	1.3	1.1	0.7	0.9	0.5	0.4		
	DB	35.9	37.7	40.5	50.7	72.9	77.1	84.7	76.9	64.1	50.4	40.0	37.9			
	MCWB	33.4	34.9	35.4	43.1	57.1	61.6	65.5	64.3	57.1	46.7	38.2	35.2			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	31.6	34.1	36.9	45.5	65.0	71.4	78.1	71.0	60.4	47.4	36.7	34.8			
	2% MCWB	30.2	32.2	33.5	38.7	51.8	57.6	63.3	61.6	54.1	44.9	35.2	33.1			
	5% DB	28.3	31.7	34.7	42.2	58.5	66.8	72.9	66.5	57.5	44.6	34.4	32.5			
	5% MCWB	27.2	30.2	31.8	36.6	48.9	55.8	61.2	59.0	52.6	42.7	33.4	31.3			
	10% DB	25.4	28.9	32.5	39.8	53.4	61.9	69.2	63.1	54.2	41.9	33.0	30.5			
	10% MCWB	24.4	27.6	30.1	35.1	45.7	53.6	60.3	57.3	50.8	40.6	32.2	29.5			
	0.4% WB	33.9	35.6	35.9	43.6	57.6	63.7	67.4	66.5	57.9	47.2	38.3	35.7			
	0.4% MCDB	36.1	36.9	38.9	51.0	71.2	73.8	74.6	74.9	61.9	49.2	39.4	37.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	30.3	32.5	34.1	39.3	52.8	60.6	65.0	62.7	55.4	45.3	35.4	33.4			
	2% MCDB	31.5	34.0	36.7	44.6	62.2	69.0	74.8	69.5	59.7	47.4	36.5	34.6			
	5% WB	27.3	30.1	32.3	37.2	49.3	57.2	63.0	60.1	53.2	43.0	33.6	31.6			
	5% MCDB	28.3	31.6	34.2	41.1	57.7	63.2	71.2	65.3	56.7	44.5	34.2	32.6			
	10% WB	24.4	27.7	30.4	35.8	46.5	54.5	61.0	57.9	51.3	40.7	32.3	29.5			
	10% MCDB	25.3	29.0	32.6	39.1	52.7	61.0	68.0	62.4	54.1	41.9	32.9	30.5			
	MDBR	18.1	17.2	20.3	17.2	16.2	15.6	16.6	16.3	13.8	9.5	12.3	16.2			
	5% DB	16.5	11.4	13.6	16.8	26.3	23.0	24.2	22.5	18.3	9.8	8.7	10.5			
(36) Mean Daily Temperature Range	MCDBR	16.1	10.1	11.1	11.1	14.2	11.3	11.7	13.3	12.5	8.0	8.3	10.1			
	MCWBR	16.1	12.1	12.3	14.9	23.4	21.0	21.8	20.1	16.6	9.2	8.3	10.4			
	MCWBR	15.7	11.0	10.5	10.3	13.3	11.6	11.5	12.5	11.6	7.7	8.1	10.1			
	MCWBR	15.7	11.0	10.5	10.3	13.3	11.6	11.5	12.5	11.6	7.7	8.1	10.1			
(40) Clear-Sky Solar Irradiance	taub	0.196	0.247	0.290	0.322	0.344	0.343	0.355	0.337	0.305	0.264	N/A	N/A			
	taud	1.913	2.058	2.145	2.217	2.393	2.476	2.464	2.492	2.546	2.261	N/A	N/A			
	Ebn at Noon	12	160	228	258	267	272	263	255	235	157	N/A	0			
	Edh at Noon	2	14	25	32	30	29	29	24	18	11	N/A	0			
(44) All-Sky Solar Radiation	RadAvg	6	134	596	1179	1466	1515	1399	983	518	164	21	0			
	RadStd	0	11	35	69	156	137	155	92	46	23	3	0			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page